

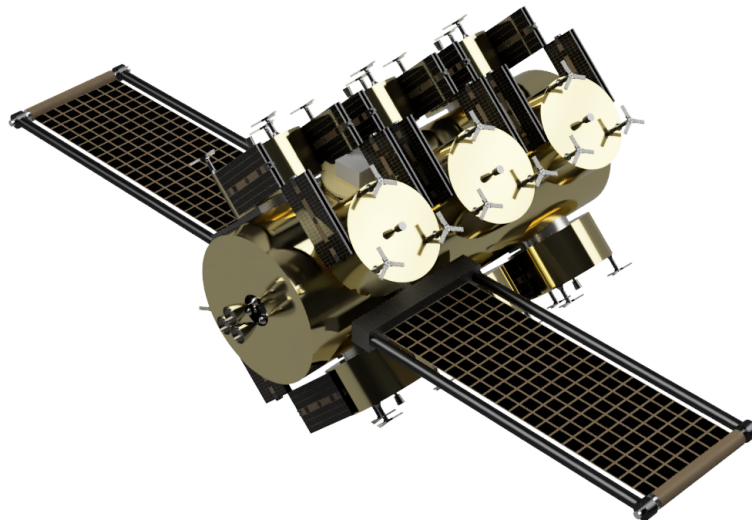
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# Design of Nonlinear Attitude Controllers via Feedback Linearization Techniques

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## **Abstract**

This report outlines the reaction control system and the attitude control laws of the drones for the *Space Gecko* mission. A combination of external thrusters and a four internal reaction wheel system capable of providing full three-axis control and stabilisation is presented. The operation of two Lyapunov-based nonlinear control laws for the attitude regulation and attitude tracking are described. Feedback linearization methods have been applied to realise a PD-like and a sliding-mode controller. Simulation results show that the proposed controllers successfully distribute the torque commands to each reaction wheel and provide the pulse-width pulse-frequency modulated signals for the operation of the on/off thruster system. The drones are able to satisfactorily perform reorientation and tracking manoeuvres and the controllers' robustness is proved with the introduction of disturbances in the system.

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# 1 Introduction

## 1.1 Motivation

Human activities in space have significantly increased the number of orbital debris, leading to a surging threat for current and future space missions. The *Space Gecko* project aims to remove objects of concern in Earth orbit with a multi-agent system, i.e. a swarm of space drones, carried by a large spacecraft known as mothership. A high-level overview of the mission profile is illustrated in Figure 1.

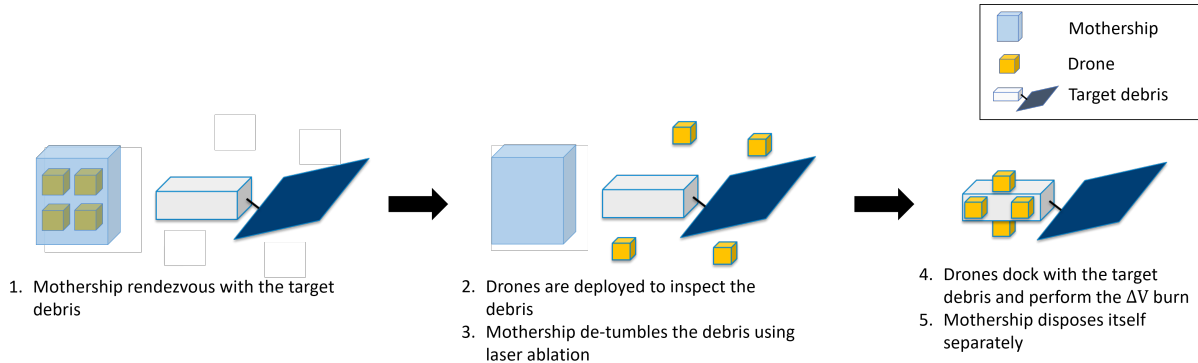


Figure 1: *Space Gecko* debris removal strategy (adapted from [1])

This report proposes a quaternion- and Lyapunov-based control law for the attitude regulation and tracking of the individual space drones, which together with a translation controller serve as the fundamental subsystems within the main inspection, attachment and deorbit controllers. The Attitude Determination and Control System (ADCS) is a crucial subsystem in a spacecraft as it provides pointing accuracy and stability of the vehicle and its payload. Being able to control the spacecraft's orientation is critical for the success of any mission, and therefore an appropriate controller must be devised to satisfy the attitude requirements. The control design presents a challenge due to the non-linear dynamics defining the rotational motion of the spacecraft. Additionally, inaccuracies in the measurements and between plant and model must be dealt with to ensure operational stability and robustness.

## 1.2 Overview

The report firstly presents the equations of motion that dictate the attitude kinematics and dynamics of a spacecraft in Section 2. Next, the control structure is explained in Section 3, namely the mothership-swarm communication topology and the drone controller architecture. The report then delves into the ADCS, initially outlining the actuators on board the spacecraft in Section 4, and later providing a detailed explanation of the feedback linearization approach for attitude regulation and the sliding-mode controller for attitude tracking in Section 5. Section 6 presents the tests conducted to simulate the system together with a discussion of the results. Finally, concluding remarks and suggestions for future iterations of the ADCS are discussed in Section 7.

# 2 Attitude Kinematics and Dynamics

The spacecraft attitude describes the rotational motion of a spacecraft relative to a defined coordinate system. Two frames of reference are introduced: the Earth-centred inertial (ECI) reference frame and the Body (B) reference frame. The former has its  $x$ -axis pointing towards the vernal equinox, the  $y$ -axis pointing  $90^\circ$  to the east of the  $x$ -axis in the equatorial plane, and the  $z$ -axis pointing northward along the Earth rotation axis [3]. The latter has its coordinate system fixed in origin to the spacecraft's centre of mass and axes pointing in the direction of the principal moments of inertia, satisfying the right-hand rule.

Different types of representations can be used for the spacecraft attitude, the most common being direction cosine matrices, Euler angles and quaternions. Quaternion representation is employed throughout this report due to its widespread use in spacecraft ADCS. Furthermore, quaternions have no singularities and are computationally less intense as any rotation can be expressed using one matrix [4].

The quaternion attitude kinematics is given by:

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{\Omega}(\boldsymbol{\omega})\mathbf{q} = \frac{1}{2} \begin{bmatrix} 0 & \omega_z & -\omega_y & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y \\ \omega_y & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} \quad (1)$$

where the scalar-last convention is used for the quaternion, i.e.  $\mathbf{q} = q_1i + q_2j + q_3k + q_4$ . The spacecraft attitude dynamics are governed by Euler's rotational equation of motion:

$$\mathbf{J}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) + \mathbf{T} \quad (2)$$

where  $\mathbf{J}$  is the spacecraft inertia tensor matrix,  $\boldsymbol{\omega}$  is the body angular velocity, and  $\mathbf{T}$  are the body torque components.

For the case where reaction wheels are used, Equation (2) becomes:

$$\bar{\mathbf{J}}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times (\bar{\mathbf{J}}\boldsymbol{\omega} + \mathbf{h}) - \dot{\mathbf{h}} \quad (3)$$

where  $\bar{\mathbf{J}}$  accounts for the inertia of the wheels,  $\mathbf{h}$  is the wheel angular momentum, and  $\dot{\mathbf{h}}$  is the wheel torque [5].

### 3 Control Structure

#### 3.1 Communication Topology

The communication structure in the mothership-swarm network effectively determines the drone's on-board computer complexity as it dictates the information received and sent by each individual drone. The aim was to minimise the number of communication channels and therefore it was decided that only a single element in the network would be responsible for the decision-making. It was natural to select the mothership as the 'brain' of the multi-agent system, hence a star-type topology was followed in the construction of the control system with the mothership as the central hub as depicted in Figure 2.

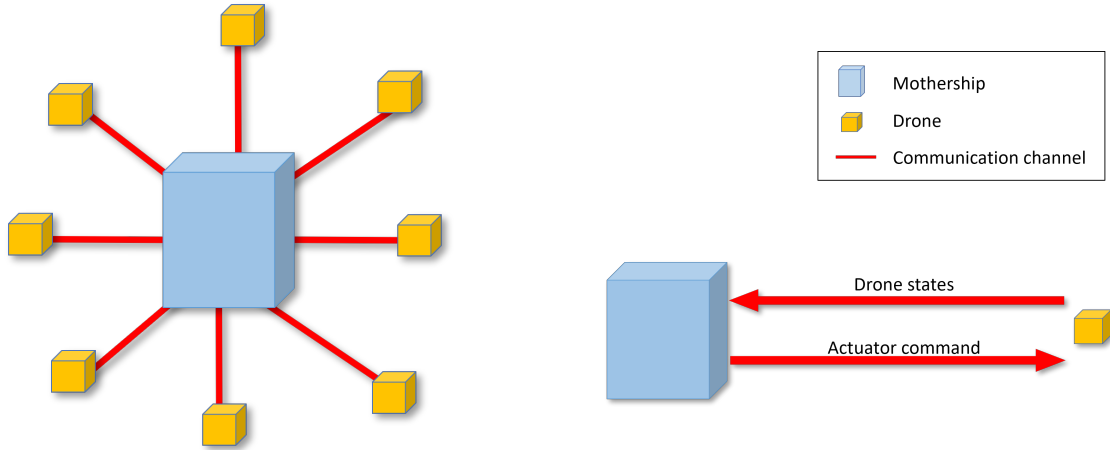


Figure 2: Communication Topology (left) and mothership and drone information exchange (right).

With this configuration, only the mothership is aware of the states (e.g. position, attitude) of the swarm. She receives the information from each agent, evaluates it against the mission requirements, and sends commands to the drones if actuation is needed. The drones are thus not autonomous in the sense that they require constant communication with the mothership to perform their duties.

#### 3.2 Control Architecture

Different controllers were developed and tailored for each distinguished mission phase and drone type. Initially, a reduced number of inspection drones are deployed to study the debris, after which the main group is released to complete the attachment and deorbit stages. The translation and attitude sub-controllers sit at the core of the main controllers, and their implementation within the main controllers is detailed in [REF PAVOL AND BLANCA]. The particular control architecture for the ADCS is illustrated in Figure 3.

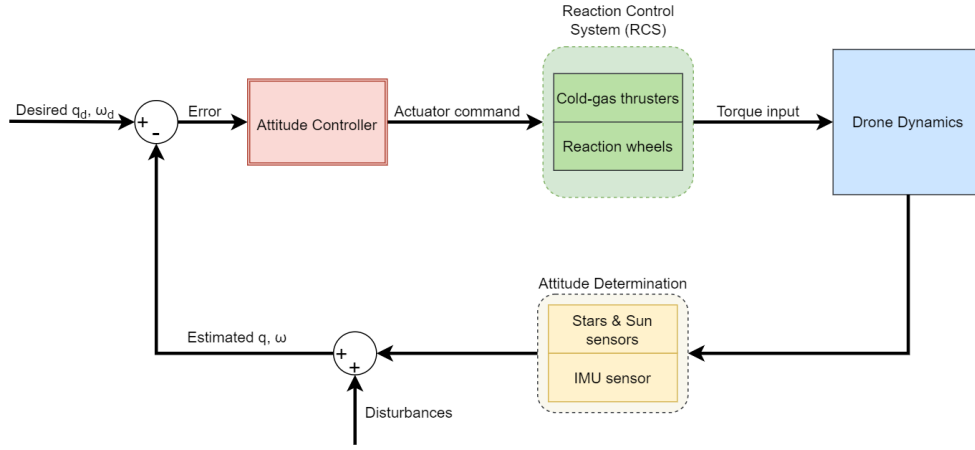


Figure 3: Attitude controller block diagram.

## 4 Reaction Control System (RCS)

A combination of internal reaction wheels and external thrusters is used to control the spacecraft's attitude. Each system is independently capable of providing three-axis stabilisation; however, depending on the task, reaction wheels might be preferred over thrusters and viceversa. The following subsections will discuss the particular uses of each subsystem in the drone.

### 4.1 Reaction Wheels

Reaction wheels (RW) are used as the main attitude control system for spacecrafts. Their advantages include high pointing accuracy and no propellant consumption which reduces the mass fraction needed for fuel. The RW system will be primarily used for attitude tracking and manoeuvres that involve high accuracy but are not time-critical, i.e. fast actuation is not vital. Four identical reaction wheels will be installed on the drones in a pyramid configuration for redundancy and safety purposes. An illustration of the pyramid configuration is shown in Figure 4.

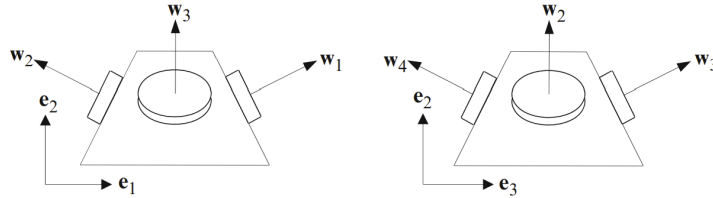


Figure 4: Four reaction wheel pyramid configuration

The main challenge is to establish a relationship between the wheel frame and the body frame. This can be obtained with a transformation matrix  $\mathbf{A}$  whose columns are unit vectors in the body frame along the spin axes of the wheels. The pyramid configuration assumes symmetry on each coordinate plane and for the case of four reaction wheels the transformation matrix is given by:

$$\mathbf{A}_4 = \begin{bmatrix} a & -a & 0 & 0 \\ b & b & c & c \\ 0 & 0 & d & -d \end{bmatrix} \quad (4)$$

where  $a^2 + b^2 = c^2 + d^2 = 1$ . This matrix has a preferential direction along the  $\mathbf{e}_2$  axis. The total wheel angular momentum  $\mathbf{h}$  in the body frame is given by:

$$\mathbf{h}_B^w = \mathbf{A}_4 \mathbf{h}_W^w \quad (5)$$

where the superscript  $w$  is the applied torque of the individual wheels. A relationship for the wheel torques can be also obtained by substituting  $\mathbf{h}$  for  $\dot{\mathbf{h}}$  in Equation (5). There is no unique way to distribute the torque or angular momentum for a four-wheel system as  $\mathbf{A}_4$  is singular, hence the Moore-Penrose pseudoinverse approach

is followed [6]. The pseudoinverse of  $\mathbf{A}_4$  is given by :

$$\mathbf{A}_4^\dagger = \frac{1}{2} \begin{bmatrix} 1/a & b/(b^2 + c^2) & 0 \\ -1/a & b/(b^2 + c^2) & 0 \\ 0 & c/(b^2 + c^2) & 1/d \\ 0 & c/(b^2 + c^2) & -1/d \end{bmatrix} \quad (6)$$

and therefore  $\mathbf{h}_W^w$  can be approximated by setting:

$$\mathbf{h}_W^w = \mathbf{A}_4^\dagger \mathbf{h}_B^w \quad (7)$$

## 4.2 Thrusters

Thrusters can generate both forces and torques and are used for translation and attitude control. The main disadvantage is that their lifespan is limited by the propellant supply. The force and torque provided by a thruster are:

$$\mathbf{F}_{thruster} = -\dot{m}\mathbf{v}_{rel} \quad (8a)$$

$$\dot{\mathbf{h}} = \mathbf{r} \times \mathbf{F}_{thruster} \quad (8b)$$

where  $\dot{m}$  is the rate at which mass is expelled,  $\mathbf{v}_{rel}$  is the velocity of the expelled mass relative to the spacecraft, and  $\mathbf{r}$  is the vector from the spacecraft center of mass to the thruster.

The thrusters will be mounted around the drones to control all degrees of rotation while a single, more powerful model will be used for deorbiting purposes. Thrusters generally provide greater torque levels than reaction wheels and do not pose the danger of saturation [7]. Thus, they will be used for manoeuvres where fast actuation is required and for spacecraft translation. In addition, they will effectively serve as a safety system to prevent or recover from wheel saturation.

## 5 Attitude Control

Section 2 presented the governing equations for rotational motion. Their nonlinear nature motivated the design of nonlinear control laws which can ensure a well-behaved and stable response. In addition, and more importantly, linear control methods that consist of linear approximations of the dynamics, e.g. Jacobian linearization, were discarded as they rely on the assumption of small range operation. The drones are designed to be agile and thus need to perform large angle manoeuvres as accurately and rapidly as possible, and for this purpose two controllers are proposed in this section. Both controllers exploit the technique of feedback linearization whereby the nonlinear system dynamics are transformed into linear systems. The first controller used for attitude regulation is essentially a PD controller, while the second is based on sliding-mode control for tracking. Lyapunov's direct method is used in both cases to construct the control laws and prove their stability, hence a brief explanation has also been included. The concept of pulse-width pulse-frequency (PWPF) modulation for the attitude control thrusters is also explored in this section.

Before delving into the controller details, some key concepts are introduced first. The error quaternion is defined by [5]:

$$\delta \mathbf{q} := \begin{bmatrix} \delta \mathbf{q}_{1:3} \\ \delta q_4 \end{bmatrix} = \mathbf{q} \otimes \mathbf{q}_d^{-1} \quad (9)$$

where

$$\delta \mathbf{q}_{1:3} = \Xi^T(\mathbf{q}_d) \mathbf{q} = \begin{bmatrix} q_4 & q_3 & -q_2 & -q_1 \\ -q_3 & q_4 & q_1 & -q_2 \\ q_2 & -q_1 & q_4 & -q_3 \end{bmatrix}_d \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} \quad (10a)$$

$$\delta q_4 = \mathbf{q}^T \mathbf{q} \quad (10b)$$

The error is quantified by the quaternion vector components, i.e.  $\delta \mathbf{q}_{1:3}$ . Therefore, the equilibrium point is defined by the identity quaternion  $\mathbf{I}_q = [0 \ 0 \ 0 \ 1]$  where  $\delta q_4 = 1$  because  $\|\mathbf{q}\| = 1$ .

### 5.1 Lyapunov Theory

Lyapunov stability theory, in particular Lyapunov's direct method, can be used to show the global asymptotic stability condition for a general nonlinear (and linear) system. The concept bases its idea on the fact that if the energy of a system is continuously dissipated, then the system must eventually settle to an equilibrium point [8]. Lyapunov stability is guaranteed if a scalar function  $V(\mathbf{x})$  with continuous first order derivatives satisfies the following conditions:

- $V(\mathbf{x}_e) = 0$  where  $\mathbf{x}_e$  is an equilibrium point
- $V(\mathbf{x}) > 0$  for  $\mathbf{x} \neq \mathbf{x}_e$
- $V(\mathbf{x}) \rightarrow \infty$  as  $\|\mathbf{x}\| \rightarrow \infty$
- $\dot{V}(\mathbf{x}) < 0$

## 5.2 Pulse-Width Pulse-Frequency (PWPF) Modulation

External-torque control requires the use of thrusters that eject mass at high speeds through a nozzle to generate angular momentum on the spacecraft. These thrusters are on/off by nature; however, the ACDS produces a continuous desired torque signal. It is therefore necessary to convert the continuous commanded torque to an on/off signal to control the thruster. The most common attitude control method by thrusters uses pulse-width pulse-frequency (PWPF) modulation due to its advantages such as pseudo-linear operation, tunability and low fuel consumption [2]. Figure 5 shows the main components of a PWPF modulator, which consist of a Schmitt trigger, a lag filter and a feedback loop.

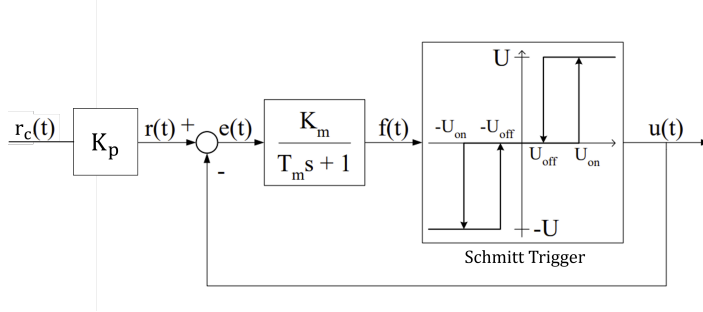


Figure 5: Pulse-width pulse-frequency system [2]

The modulator has different parameters that need to be adjusted depending on the system. The variable  $K_p$  is a gain used to amplify or reduce the commanded input. The filter is a first-order filter with time constant  $T_m$  and gain  $K_m$ . Table 1 from [2] offers a guidance on recommended parameter values. Further information on the theory behind the operation of the PWPF modulator such as on- and off-time, frequency and duty cycle can also be found in this reference.

Table 1: Recommended PWPF modulator parameters.

| Parameter   | $K_m$         | $T_m$             | $U_{on}$       | $U_{off}$             | $K_p$         |
|-------------|---------------|-------------------|----------------|-----------------------|---------------|
| Recommended | $2 < K_m < 7$ | $0.1 < T_m < 0.5$ | $U_{on} > 0.3$ | $U_{off} < 0.8U_{on}$ | $K_p \geq 20$ |

## 5.3 Attitude Regulation

Regulation control is defined as rotating the spacecraft to a desired fixed orientation. The objective is to achieve a constant attitude given by  $\mathbf{q}_d$ , meaning that the angular velocity must become zero at the end of the manoeuvre.

The controller is designed such that the spacecraft is capable of performing arbitrary large angle reorientation manoeuvres. To this end, several quaternion-based feedback control laws have been proposed in [9]. The following feedback linearization has been implemented for the attitude regulation:

$$\mathbf{u} = -k_p \delta \mathbf{q}_{1:3} - k_d \boldsymbol{\omega} \quad (11)$$

where  $\mathbf{u} \equiv \mathbf{T}$  in Equation (2) and  $k_p$  and  $k_d$  are the *proportional* and *derivative* gains, respectively. Thus, the described control law is effectively equivalent to a PD controller. Substituting Equation (11) into (2) gives the closed-loop dynamics:

$$\dot{\boldsymbol{\omega}} = -J^{-1} [\boldsymbol{\omega} \times (J\boldsymbol{\omega}) + k_p \delta \mathbf{q}_{1:3} + k_d \boldsymbol{\omega}] \quad (12)$$

From this result it is observed that the only equilibrium point is  $\mathbf{x}_{eq} = [\mathbf{q}_{1:3}^T \quad \boldsymbol{\omega}^T]^T = [\mathbf{0} \quad \mathbf{0}]^T$ . The proposed attitude regulation controller can also be applied to a reaction wheel control system by rewriting Equation 3 in an alternative form:

$$\bar{J}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times (\bar{J}\boldsymbol{\omega}) + \bar{\mathbf{u}} \quad (13a)$$

$$\dot{\mathbf{h}} = -\boldsymbol{\omega} \times \mathbf{h} - \bar{\mathbf{u}} \quad (13b)$$

where  $\bar{\mathbf{u}} \equiv \mathbf{u}$  is the effective wheel torque input.

The asymptotic stable response can be demonstrated with Lyapunov's direct method. Consider the following Lyapunov function from [5]:

$$V = \frac{1}{4}\boldsymbol{\omega}^T J \boldsymbol{\omega} + \frac{1}{2}k_p \delta \mathbf{q}_{1:3}^T \delta \mathbf{q}_{1:3} + \frac{1}{2}k_p(1 - \delta q_4)^2 \quad (14)$$

which satisfies the three first conditions. The time derivative is given by:

$$\dot{V} = \frac{1}{2}\boldsymbol{\omega}^T J \dot{\boldsymbol{\omega}} + k_p \delta \mathbf{q}_{1:3}^T \delta \dot{\mathbf{q}}_{1:3} - k_p(1 - \delta q_4)\delta \dot{q}_4 \quad (15)$$

From Equation (1), the time derivative for the error quaternion can be derived as:

$$\delta \dot{\mathbf{q}}_{1:3} = \frac{1}{2}\delta \mathbf{q}_{1:3} \times \boldsymbol{\omega} + \frac{1}{2}\delta q_4 \boldsymbol{\omega} \quad (16a)$$

$$\delta \dot{q}_4 = -\frac{1}{2}\delta \mathbf{q}_{1:3}^T \boldsymbol{\omega} \quad (16b)$$

Substituting these results together with Equation (12) into Equation (15) leads to:

$$\begin{aligned} \dot{V} &= -\frac{1}{2}(\boldsymbol{\omega}^T \delta \mathbf{q}_{1:3} [k_p + k_p \delta q_4 - k_p(1 - \delta q_4)] - \frac{1}{2}k_d \boldsymbol{\omega}^T \boldsymbol{\omega}) \\ &= -\frac{1}{2}k_d \boldsymbol{\omega}^T \boldsymbol{\omega} \end{aligned} \quad (17)$$

which is clearly negative definite and hence satisfies the last condition for Lyapunov stability. A final remark is that it is always desired to provide the shortest path to the desired attitude due to power and energy consumption purposes. To achieve this, an extra factor is added to the control law in Equation (11) as proven in [10]:

$$\mathbf{u} = -k_p \text{sign}(\delta q_4) \delta \mathbf{q}_{1:3} - k_d \boldsymbol{\omega} \quad (18)$$

If  $\delta q_4 < 0$ , a positive feedback term is introduced, which provides the shorter path to reach the desired point.

## 5.4 Attitude Tracking

The approach in the previous section allowed the spacecraft to perform reorientation manoeuvres to a fixed objective attitude while driving the angular velocity to zero. However, it is often desired and necessary to follow time-varying trajectories such as inspection manoeuvres. An attitude tracking controller based on Sliding-Mode Control (SMC) is described in this section.

### 5.4.1 Sliding-Mode Control Theory

SMC is a variable structure control for nonlinear systems. The term *variable structure control* (VSC) refers to a form of discontinuous control whereby the dynamics of the system are altered due to the switching between control laws based on the position of the state trajectory. The strengths of SMC and their proof can be found in [8] and [11]. Some of the advantages include:

- SMC belongs to the so-called 'robust' controllers because it has low sensitivity against uncertainties, disturbances and modeling imprecisions,
- it introduces a notational simplification where  $n^{\text{th}}$ -order problems are effectively reduced to 1<sup>st</sup>-order problems, and
- convergence to the desired state is achieved in finite-time.

Finite-time convergence is achieved by first choosing a candidate Lyapunov function so that the state trajectories of the system remain stable and point toward the origin. The switching control law forces the state trajectories onto a surface in the phase plane called the *sliding surface* in a process known as *reaching phase*. Once the sliding surface is reached, the controller forces the state to slide towards the desired state in a process known as *sliding phase* or *sliding mode*.

Consider the following single-input single-output system:

$$\mathbf{x}^n = f(\mathbf{x}) + u \quad (19)$$

where  $f(\mathbf{x})$  may be a linear or nonlinear function of the state vector  $\mathbf{x} = [x \ \dot{x} \ \dots \ x^{(n-1)}]^T$  and  $u$  is the input. Let  $\mathbf{x}_d = [x_d \ \dot{x}_d \ \dots \ x_d^{(n-1)}]^T$  be the desired state and  $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_d$  be the tracking error vector. The control problem is to get the state  $\mathbf{x}$  to track  $\mathbf{x}_d$  in the presence of modelling imprecision on  $f(\mathbf{x})$ . Now define the sliding surface  $s$  as:

$$s = \left( \frac{d}{dt} + \lambda \right)^{n-1} \tilde{x} \quad (20)$$

where  $\lambda$  is a strictly positive constant. The sliding surface can essentially reduce any system to a 1<sup>st</sup>-order system by a form of feedback linearization. For  $n = 2$ , Equation (20) becomes:

$$s = \dot{\tilde{x}} + \lambda \tilde{x} \quad (21)$$

Consider the following candidate Lyapunov function and its time derivative:

$$V(\tilde{x}) = \frac{1}{2} s^2 \quad (22a)$$

$$\dot{V}(\tilde{x}) = s \dot{s} \quad (22b)$$

From these a control law can be derived such that the state trajectories are asymptotically stable.

The problem of plant and model mismatch is now addressed. Typically, the spacecraft system dynamics are not exactly known and therefore uncertainties must be considered to ensure robustness. Consider the following 2<sup>nd</sup>-order system:

$$\ddot{x} = \bar{f}(x, \dot{x}) + u \quad (23)$$

where  $\bar{f}(x, \dot{x})$  is a model of the actual dynamics  $f(x, \dot{x})$ . Taking the time derivative of Equation (21) and substituting Equation (23) gives:

$$\dot{s} = \ddot{\tilde{x}} + \lambda \dot{\tilde{x}} = \bar{f}(x, \dot{x}) + u - \ddot{x}_d + \lambda \dot{\tilde{x}} \quad (24)$$

Motion off the sliding surface is prevented by setting  $\dot{s} = 0$ , leading to the following equivalent control law:

$$u_{eq} = -\bar{f}(x, \dot{x}) + \ddot{x}_d - \lambda \dot{\tilde{x}} \quad (25)$$

where  $u_{eq}$  is the continuous control law that would maintain  $\dot{s} = 0$  if the dynamics were exactly known [8]. In order to account for model uncertainties a discontinuous term is added across the sliding surface:

$$u = u_{eq} - k \text{sign}(s) \quad (26)$$

for some scalar  $k$ . Substituting this control law in Equation (24) yields:

$$\dot{s} = -k \text{sign}(s) \quad (27)$$

and substituting this result into Equation (22b) gives:

$$\dot{V}(\tilde{x}) = [f(x, \dot{x}) - \bar{f}(x, \dot{x}) - k \text{sign}(s)] s = [f(x, \dot{x}) - \bar{f}(x, \dot{x})] s - k|s| \quad (28)$$

Assume now that the model error is bounded, i.e. that  $|f(x, \dot{x}) - \bar{f}(x, \dot{x})| \leq F_{max}$  and set  $k = F_{max} + \mu$  for some positive scalar  $\mu$ . Equation (28) becomes:

$$\dot{V}(\tilde{x}) \leq -\mu|s| \quad (29)$$

which proves that the proposed control law is stable in face of modeling uncertainties if  $k$  is large enough.

One drawback of the discontinuous control law is the ‘chattering’ phenomenon caused by the inevitable imperfect implementation of the control switchings, e.g. switching is not instantaneous. Chattering is undesired since it involves high control activity, hence the control law is smoothed to achieve an optimal trade-off between control bandwidth and tracking precision. A proposed solution in [8] consists in defining a boundary layer close to the sliding surface where the signum function in Equation (26) is replaced with a saturation function:

$$u = u_{eq} - k \text{sat}(s, \epsilon) \quad (30)$$

where  $\epsilon$  is the boundary layer thickness and

$$\text{sat}(s, \epsilon) \equiv \begin{cases} 1 & \text{for } s_i > \epsilon_i \\ s_i/\epsilon_i & \text{for } |s_i| \leq \epsilon_i \\ -1 & \text{for } s_i < -\epsilon_i \end{cases} \quad (31)$$

The tracking performance is suboptimal with the saturation function compared to the signum function. However, it provides an adequate balance between robustness and practical implementation for a smoothed control input.

### 5.4.2 Implementation

A sliding-mode controller based on the results from the previous section and [10] has been implemented for the spacecraft attitude tracking. Following Equation (21), the sliding surface vector is given by:

$$\mathbf{s} = (\boldsymbol{\omega} - \boldsymbol{\omega}_d) + k\delta\mathbf{q}_{1:3} \quad (32)$$

where  $\boldsymbol{\omega}_d$  is a desired angular velocity. Note that the structure of the sliding surface is similar to the attitude regulation control law and that attitude tracking  $\equiv$  regulation when  $\boldsymbol{\omega}_d = 0$ . Taking the time derivative of Equation (32) gives:

$$\dot{\mathbf{s}} = (\dot{\boldsymbol{\omega}} - \dot{\boldsymbol{\omega}}_d) + k\delta\dot{\mathbf{q}}_{1:3} \quad (33)$$

where the time derivative of the error quaternion is [12]:

$$\delta\dot{\mathbf{q}}_{1:3} = \frac{1}{2}\delta\mathbf{q}_{1:3} \times (\boldsymbol{\omega} + \boldsymbol{\omega}_d) + \frac{1}{2}\delta q_4(\boldsymbol{\omega} - \boldsymbol{\omega}_d) \quad (34)$$

Substituting this result together with the expression for the angular acceleration from Equation (2) into Equation (33) leads to:

$$\dot{\mathbf{s}} = -J^{-1}[\boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) + \mathbf{T}_{eq}] - \dot{\boldsymbol{\omega}}_d + \frac{k}{2}[\delta q_4(\boldsymbol{\omega} - \boldsymbol{\omega}_d) + \delta\mathbf{q}_{1:3} \times (\boldsymbol{\omega} + \boldsymbol{\omega}_d)] \quad (35)$$

where  $\mathbf{T}_{eq} \equiv \mathbf{u}_{eq}$  because the equivalent control law must first be derived. By setting Equation (35) to zero:

$$\mathbf{u}_{eq} = J \left\{ \frac{k}{2}[\delta q_4(\boldsymbol{\omega}_d - \boldsymbol{\omega}) - \delta\mathbf{q}_{1:3} \times (\boldsymbol{\omega} + \boldsymbol{\omega}_d)] + \dot{\boldsymbol{\omega}}_d \right\} + \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) \quad (36)$$

the equivalent control law is obtained. The discontinuous term is then added to account for model uncertainties in the form of a saturation function to minimise chattering. Furthermore, and similar to the regulation case, a  $\text{sign}(\delta q_4)$  factor is also added to guarantee that the spacecraft follows the shortest distance to the objective. These modifications lead to the following sliding surface and sliding-mode control law:

$$\mathbf{s} = (\boldsymbol{\omega} - \boldsymbol{\omega}_d) + k\text{sign}(\delta q_4)\delta\mathbf{q}_{1:3} \quad (37a)$$

$$\mathbf{u} = J \left\{ \frac{k}{2}[\delta q_4(\boldsymbol{\omega}_d - \boldsymbol{\omega}) - \text{sign}(\delta q_4)\delta\mathbf{q}_{1:3} \times (\boldsymbol{\omega} + \boldsymbol{\omega}_d)] + \dot{\boldsymbol{\omega}}_d - G\bar{\mathbf{s}} \right\} + \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) \quad (37b)$$

where  $G$  is a positive definite gain matrix and the  $i$ -th component of  $\bar{\mathbf{s}}$  is given by:

$$\bar{s}_i = \text{sat}(s_i, \epsilon_i), \quad i = 1, 2, 3 \quad (38)$$

### 5.4.3 Robustness

The robustness of the proposed variable structure controller is assessed in this section as per [10]. Firstly, define the following bounded modelling errors for the inertia matrix:

$$J^{-1} = \hat{J}^{-1} + \delta J^{-1} \quad (39)$$

where  $\hat{J}$  is the nominal inertia matrix. Neglecting the nonlinear term in Equation 2 and adding an external (bounded) disturbance  $\mathbf{d}$  yields:

$$\dot{\boldsymbol{\omega}} = J^{-1}\mathbf{u} + J^{-1}\mathbf{d} \quad (40)$$

The time derivative of the sliding surface can therefore be approximated by:

$$\dot{\mathbf{s}} \approx \delta J^{-1}\hat{J}\mathbf{u} - J^{-1}\hat{J}G\bar{\mathbf{s}} + J^{-1}\mathbf{d} \quad (41)$$

where the higher-order perturbations in the inertia matrix are assumed to be small. Further assume that the boundary layer thickness  $\epsilon$  is sufficiently small and that the gain  $G$  is sufficiently large to keep the time derivative of the associated Lyapunov function ( $V = \frac{1}{2}\mathbf{s}^T J\mathbf{s}$ ) negative definite with modeling errors and external disturbances. The dynamics of the sliding surface are thus approximated by:

$$\dot{\mathbf{s}} \approx -\frac{1}{\epsilon}J^{-1}\hat{J}G\mathbf{s} + J^{-1}\mathbf{d} \quad (42)$$

If the sliding function settles to a finite constant  $\mathbf{s}_{ss}$ , the steady-state value satisfies the following inequality:

$$\|\mathbf{s}_{ss}\| \leq \left\| \epsilon \left( \hat{J}G \right)^{-1} \right\| \|\mathbf{d}\|_{max} \quad (43)$$

## 6 Simulation and Testing

The controllers presented in the previous sections, namely Equation (18) for regulation and Equation (37b) for tracking, have been implemented and tested on Simulink<sup>®</sup>. The same control law has been used for the three rotational degrees of freedom, i.e. roll, pitch and yaw. To simulate the behaviour of the space drones, the Spacecraft Dynamics block within the Aerospace Blockset<sup>™</sup> was used. Arbitrary gains are present in both controllers and hence tuning was required to achieve the desired performance. An overview of the tuning process that led to the final gain values is provided in Appendix A. The parameters for the PWPF modulator were selected based on the guidance from Table 1, and the values used for the simulations are:

Table 2: PWPF modulator simulation parameters.

| Parameter | $K_m$ | $T_m$ | $U_{on}$ | $U_{off}$   | $K_p$ |
|-----------|-------|-------|----------|-------------|-------|
| Value     | 5     | 0.5   | 0.7      | $0.6U_{on}$ | 30    |

### 6.1 Regulation Controller Results

The parameters for the regulation case are given in Table 3. Figure 6a shows that the reaction wheels satisfactorily reorient the drone to the identity quaternion from an arbitrary initial condition  $\mathbf{q}_0$ . The steady state is reached after approximately 2 minutes and the angular velocity is driven to zero at the end of the manoeuvre, as observed in Figure 6b. The subscripts  $x$ ,  $y$  and  $z$  indicate roll, pitch and yaw in the spacecraft body frame, respectively. Figure 6c shows the torque commands where the values in the transformation matrix from Equation 4 are  $a = b = c = d = 1/\sqrt{2}$ . Additionally, bounds of  $|\dot{\mathbf{h}}| < 0.15$  were applied to respect the physical limits of the reaction wheels.

Table 3: Attitude regulation controller parameters.

| Parameter | $k_p$ | $k_d$ | $\mathbf{q}_0$                           | $\mathbf{q}_d$      | $\boldsymbol{\omega}_0$ (deg/s) | $\boldsymbol{\omega}_d$ (deg/s) |
|-----------|-------|-------|------------------------------------------|---------------------|---------------------------------|---------------------------------|
| Value     | 2.13  | 8.24  | $[0.4939 \ 0.0994 \ 0.8256 \ -0.2541]^T$ | $[0 \ 0 \ 0 \ 1]^T$ | $[0 \ 0 \ 0]^T$                 | $[0 \ 0 \ 0]^T$                 |

When fast rotation is required, the thrusters are able to rapidly provide the external torque to perform the attitude change. The converted continuous signal to on/off commands is shown in Figure 6d. With the use of thrusters, the drone is able to reach the desired attitude in less than 1 minute, greatly reducing the manoeuvre time compared to when reaction wheels.

### 6.2 Tracking Controller Results

The parameters for the attitude tracking controller are given in Table 4. The simulation results in Figure 7a show that the drone achieves the desired angular velocity in 20 seconds and it is capable of maintaining the speed for the duration of the simulation. In addition to the quaternion representation for the attitude variation given in Figure 7b, the change of the Euler angles with time has been also included (Figure 7c) as it provides a more intuitive picture of the system dynamics. It can be observed that a 180 degrees rotation around the  $z$ -body axis is performed after a little over 180 seconds, which complies with the desired 1 deg/s velocity. The roll and pitch angles remain constant at zero degrees as per the requirements of the simulation, albeit minimal oscillations can be noticed. Figure 7d shows that only two of the reaction wheels are required to perform the rotation. Note that the wheel torque limits are not exceeded to avoid saturation of the reaction wheels.

Table 4: Attitude regulation controller parameters.

| Parameter | $k$                    | $G$                              | $\epsilon$ | $\mathbf{q}_0$  | $\boldsymbol{\omega}_0$ (deg/s) | $\boldsymbol{\omega}_d$ (deg/s) |
|-----------|------------------------|----------------------------------|------------|-----------------|---------------------------------|---------------------------------|
| Value     | $1.745 \times 10^{-4}$ | $0.015(\mathbf{I}_{3 \times 3})$ | 0.05       | $[0 \ 0 \ 0]^T$ | $[0 \ 0 \ 0]^T$                 | $[0 \ 0 \ 1]^T$                 |

The dynamics of the sliding surface can be seen in the phase plot in Figure 8a. The reaching and sliding phases are clearly distinguished: the system that starts at zero angular velocity reaches the desired yaw rate. Once there, it slides along  $s = 0$  to track the continuously varying desired yaw angle. It is worth noting that the angular velocity is not tracked perfectly but with an error smaller than  $5 \times 10^{-4}$ . This is coherent with the expected behaviour as the continuous approximation with the saturation function cannot guarantee perfect tracking but only within a precision smaller than the boundary layer thickness. The saturation significantly smooths the control discontinuity inside the boundary layer as it essentially assigns a lowpass filter structure to the local dynamics of the variable  $s$  [8]. This result contrasts with Figure 8b, where the signum function

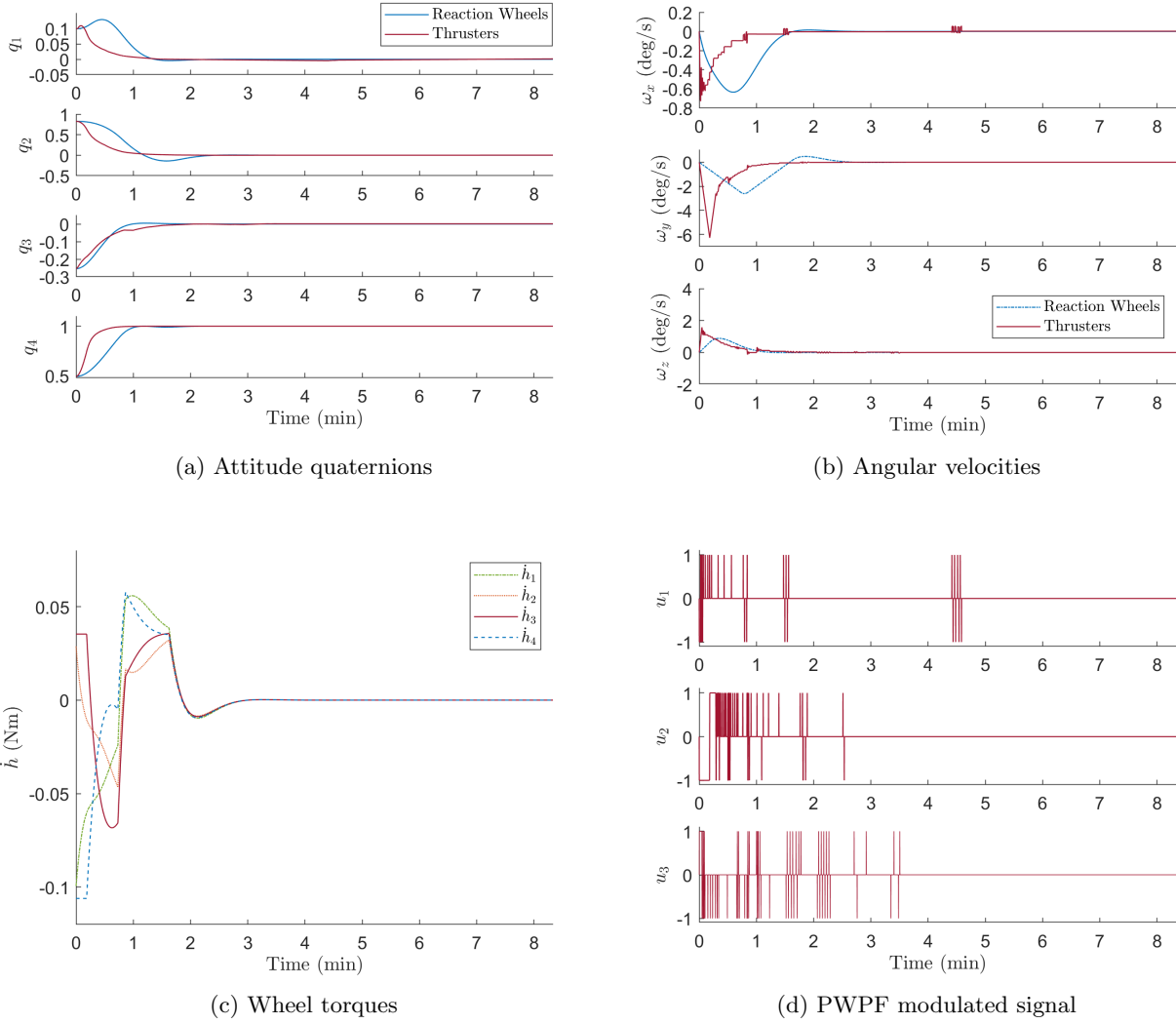


Figure 6: Simulation results for attitude regulation

has been used instead. After reaching the sliding surface, the imperfect control switchings lead to high control activity and chattering which is clearly undesirable in any system.

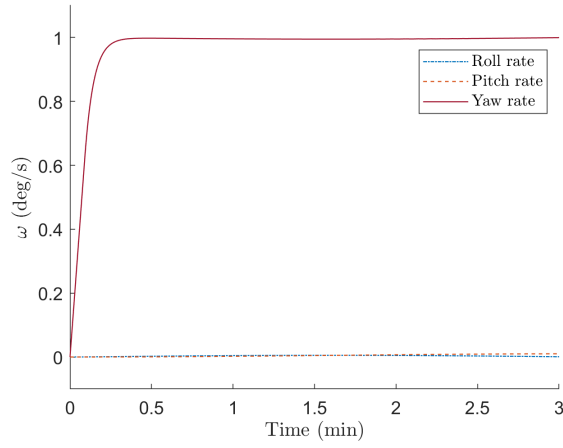
The angular velocity tracking performance of the two SMC versions has also been studied. When the signum function is used, the desired velocity is achieved in approximately 8 seconds compared to the 20 seconds taken with the saturation function as seen in Figure 9a. The chattering phenomenon can be clearly observed in Figure 9b where the error oscillates intensely with the discontinuous control law. The tracking performance in both cases is very similar due to the small value chosen for the boundary layer.

### 6.3 Robustness Study

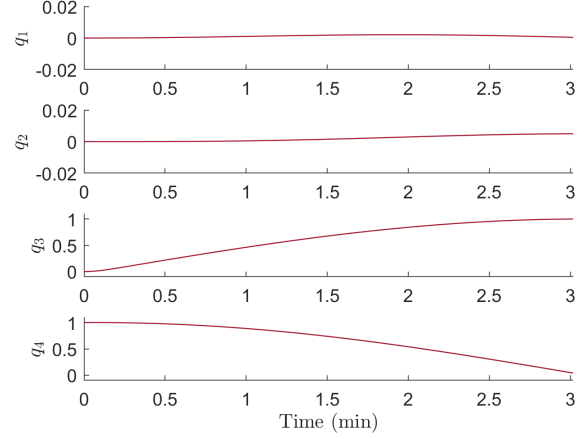
The controllers' performance against undesired inputs into the system was evaluated. The following torque disturbances were added:

$$\mathbf{d} = [0.05\cos(0.05t) \quad 0.02 \quad 0.05\sin(0.05t)]^T \text{ Nm}$$

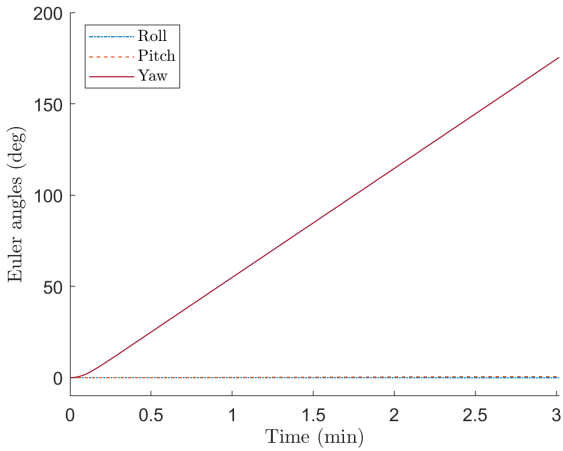
which are much higher than the expected by satellites in Low-Earth Orbit [13]. For the regulation case the drones are able to complete the desired rotation as observed in Figure 10a. However, the reaction wheels are not able to eliminate the 0.04 steady state error. As expected, the error between the actual and the desired attitude decays significantly faster with thrusters compared to reaction wheels. Figure 10b shows that the sliding function trajectory is bounded in the expected maximum bound as given by Equation (43). Since the sliding surface is a weighted sum of the position and velocity errors, it is proven that the SMC is capable of maintaining the error in the order of  $10^{-3}$  even when disturbances are present.



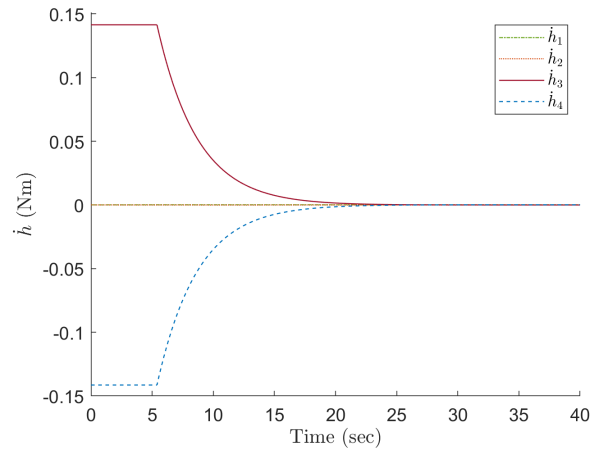
(a) Angular velocities



(b) Attitude quaternions



(c) Attitude Euler angles



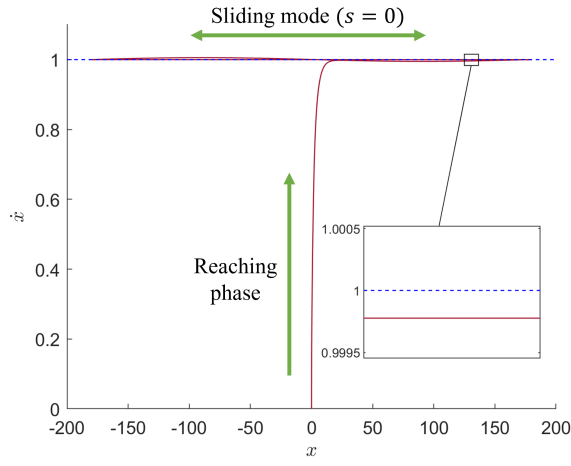
(d) Wheel torques

Figure 7: Simulation results for attitude tracking

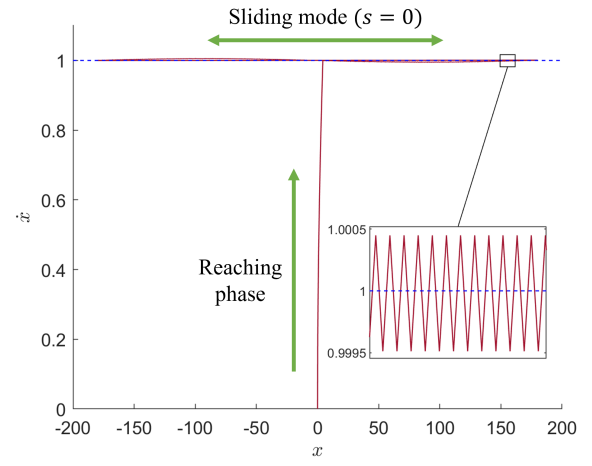
## 7 Conclusion

Two nonlinear controllers based on feedback linearization techniques have been realised for the attitude tracking and regulation of the drones that conform the swarm in the *Space Gecko* mission. The proposed control laws satisfy Lyapunov's stability condition and thus ensure asymptotic stability. The simulations indicate that the PD controller for attitude regulation enables the spacecraft to perform full three-axis reorientation manoeuvres with great precision. In the presence of undesired torques, however, the reaction wheels alone cannot eliminate the steady state error in contrast with the thrusters which are able to minimise the effect of disturbances. As expected, the use of external torques reposition the spacecraft faster than the internal reaction wheels. For the case of time-varying states, the variable structure controller successfully tracks the desired angular positions and velocities. The continuous approximation of the switching law greatly reduces chattering and control effort and barely affects the tracking performance. The robust nature of sliding controllers is demonstrated in the simulations when disturbances are introduced in the system.

This report serves as a preliminary study for the attitude control system on board the drones. Further tests should be conducted to obtain a more complete picture of the system behaviour under varying circumstances. The steady state error observed in the disturbed attitude regulation case, especially under reaction wheels actuation, could be eliminated including and integral term in the proposed PD-like controller. Regarding the slide-mode control, the trade-off between tracking precision and robustness could be further investigated. For example, the boundary layer thickness  $\epsilon$  could be made time-varying to exploit the control bandwidth. A state estimator such as a Kalman filter could be included to further protect the controller against uncertainties and attitude determination inaccuracies. Finally, the actuation of the reaction control system could be designed in such a way that the thrusters and reaction wheels operate in unison to minimise power/propellant consumption, avoid reaction wheel saturation, and maximise pointing accuracy.

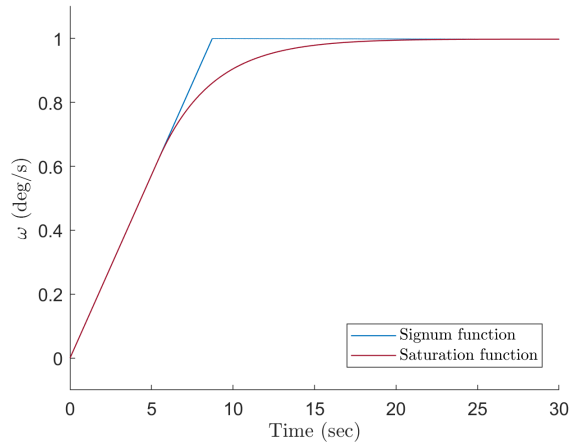


(a) Smoothed control with saturation function

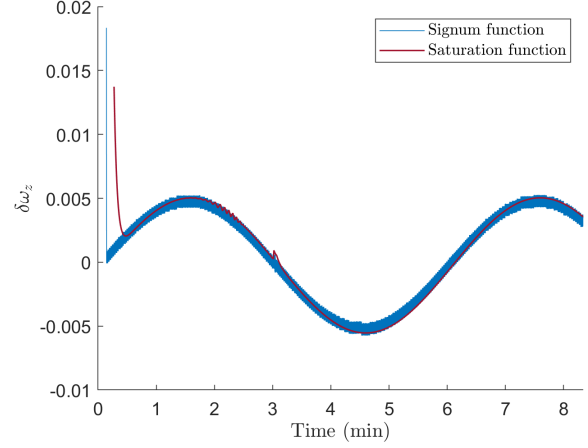


(b) Chattering phenomenon with signum function

Figure 8: Phase portraits

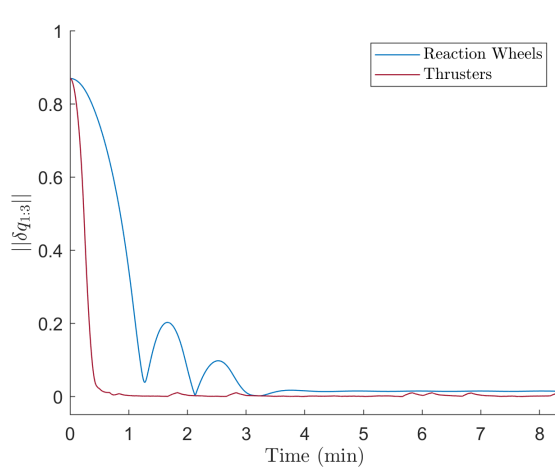


(a) Angular velocity

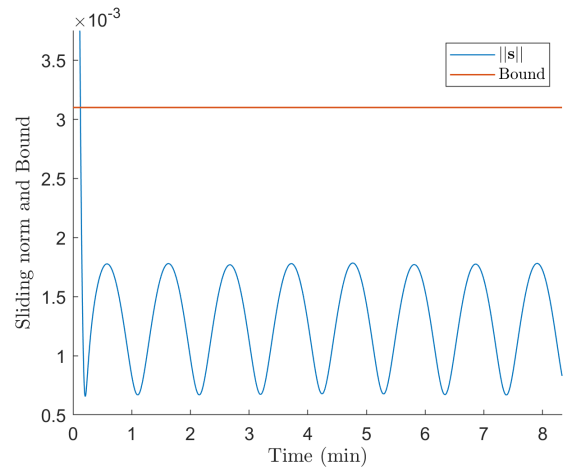


(b) Tracking error

Figure 9: Comparison of simulation results for attitude tracking with signum and saturation functions



(a) Attitude regulation quaternion error norm



(b) Sliding surface norm and bound

Figure 10: Simulation results with disturbance torques

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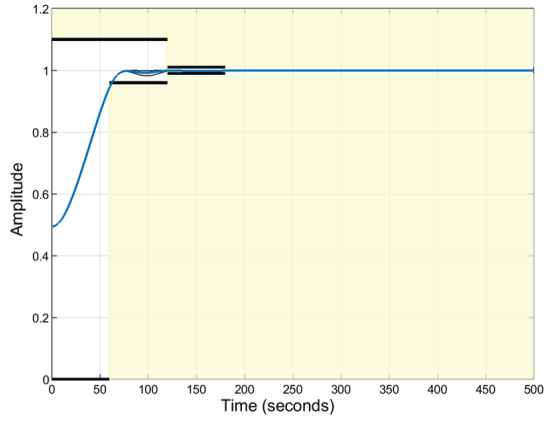
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## Appendix A Tuning of Gains

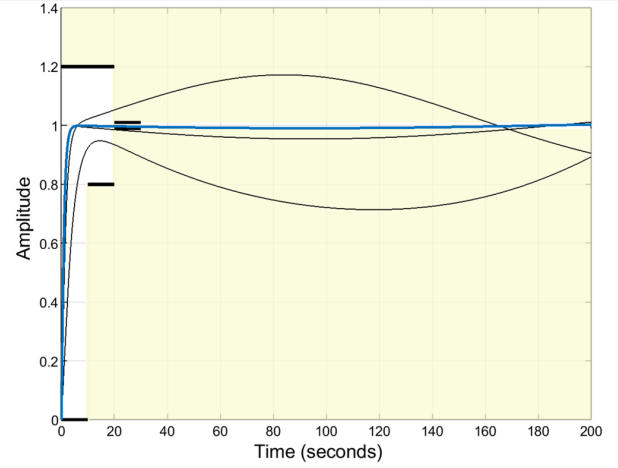
The Response Optimizer tool available on Simulink<sup>®</sup> was used to find the gains in Equations (18) and (37b). The *Check Step Response Characteristics* block allowed the definition of the step response characteristics. The constraints imposed on the step response are given in Table 5. For the regulation case, the optimization was conducted on the response characteristics of  $q_4$  while for tracking the optimization was conducted on  $\omega_z$ . The optimization history can be seen in Figure 11. The final gain values determined by the optimizer can be found in Tables 3 and 4.

Table 5: Step response constraints

|                     | Regulation | Tracking |
|---------------------|------------|----------|
| Step time (sec)     | 0          | 0        |
| Initial value       | 0.4939     | 0        |
| Final value         | 1          | 1        |
| Rise time (sec)     | 60         | 30       |
| % Rise              | 96         | 90       |
| Settling time (sec) | 0          | 60       |
| % Settling          | 1          | 1        |
| % Overshoot         | 20         | 20       |
| % Undershoot        | 0          | 0        |



(a) Regulation



(b) Tracking

Figure 11: Step response optimizer history